

Smart City Citizen Portal

Programming Fundamentals — Java Project Description

■ Project Description

This is a **Console-Based Java Application** that works as a digital portal for citizens and administrators of a Smart City. Citizens can register, login and use city services like submitting complaints, paying bills, booking appointments and viewing announcements. Admin can monitor and manage everything. All data is **permanently saved in text files** so it remains safe even after the program closes.

■ Purpose — What Is It Used For?

Feature	Description
■ Registration & Login	Citizens create accounts using their CNIC
■ Complaint System	Submit complaints about Electricity, Water, Roads
■ Utility Bill	Check and pay utility bills
■ Announcements	View latest city news and notices
■ Emergency Contacts	Access Police, Ambulance, Fire Brigade numbers
■ Appointment Booking	Book appointments at NADRA, Hospital, etc.
■ Lost & Found	Report or search for lost items
■ Feedback System	Rate and review city services
■ Admin Panel	Admin manages and monitors everything

■ Algorithm — Step by Step

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PROGRAM STARTS
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+-- STEP 1: Load all saved data from files
|
+-- STEP 2: Show Main Menu
|
+-- Register --> Validate input --> Save to file
|
+-- Citizen Login --> Open Citizen Menu
|   +-- Submit Complaint
|   +-- Pay Utility Bill
|   +-- Book Appointment
|   +-- View Announcements
|   +-- Lost & Found
|   +-- Emergency Contacts
|   +-- Give Feedback
|
+-- Admin Login --> Open Admin Menu
|   +-- View & Update Complaints
|   +-- View All Citizens
|   +-- View Appointments
|   +-- View Feedback & Ratings
|
+-- Exit --> Save all data to files
|
PROGRAM ENDS
```

■ Java Topics Used

#	Topic	Purpose
1	File Handling	Save and load all data permanently using .txt files
2	ArrayList	Store dynamic data like citizens, complaints, appointments
3	Loops	Repeat menus, read files line by line, search through lists
4	Conditionals	Handle menu choices, input validation, and searching
5	String Methods	Compare, split, search and validate user input
6	Static Methods	Call functions directly without creating objects
7	Exception Handling	Handle file errors using try-catch
8	Type Conversion	Convert file data from String to int / boolean
9	Arrays	Store fixed data like bills and announcements
10	Regex	Validate CNIC format

■ Files Created by the Program

File	Data Stored
citizens.txt	Registered citizen information
complaints.txt	Complaints and their status
appointments.txt	Booked appointments
lostfound.txt	Lost and found posts
feedback.txt	Service ratings
bills.txt	Bill payment status